Case Study: Autonomous Operations Analysis

Vehicle Utilization, Manual Intervention, Headcount Forecast

Daniel Yang | **Period:** Q1 2017 Historical + Q2 Forecast

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CONTENT

01 **Fleet Overview**
General data summary

02 **Vehicle Utilization**
Location split, total hours, speed profiles & avg duration

03 **Manual Intervention Analysis**
MPI by software (AFX12 to AFX19), location & all 18 vehicles

04 **Fleet Health by Vehicle**
Full MPI ranking across 18 vehicles — location color-coded

05 **Speed Profile & Time-of-Day**
Route mix, hourly MPI patterns & afternoon scheduling advantage

06 **Vehicle Reliability & Troubleshooting Rate**
Trouble-rate tiers, the V18 paradox & maintenance priorities

07 **Recommendations**
3 options to increase throughput with pros & risks

08 **Additional Analyses Needed**
Data gaps, open questions & future analysis priorities

09 **Headcount Forecast — Methodology**
ETS model, OT-solver assumptions & two-method design

10 **Staffing Methods A vs B — Baseline**
Weeks 1–13 comparison: location-based vs vehicle-level

11 **Non-OT Staffing & Cost Forecast**
Peak ~109 vs 98 FTE · 13-wk cost ~$2.22M · hiring-led plan

12 **With-OT Staffing & Cost Forecast**
Peak ~107 vs 83 FTE · 13-wk cost ~$1.80M · flexible plan

13 **Forecast Method Comparison**
Location-based planning vs vehicle-level scheduling precision

Index
Summary & Next Steps, Daily Performance Trends (Q1 2017), Vehicle Level Operator Hours Forecast

Part I.
Operational Performance Analysis

FLEET OVERVIEW

3,720

Total Tests Run

18

Active Vehicles

2

Locations (A & B)

317,569

Total Miles Driven

995

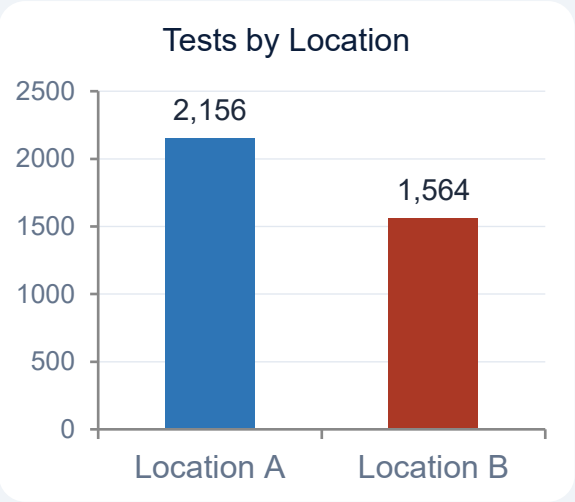
Manual Interventions

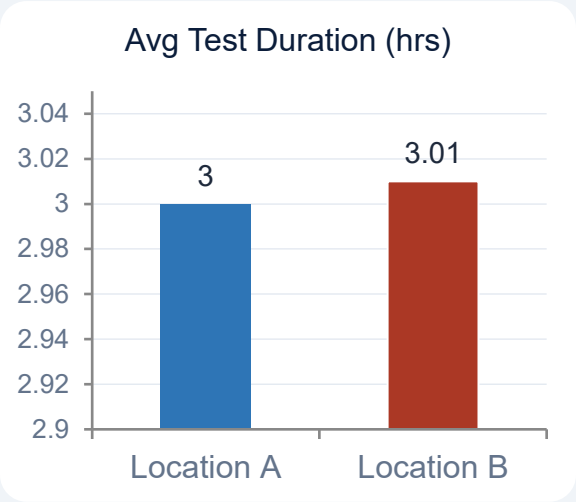
319 mi

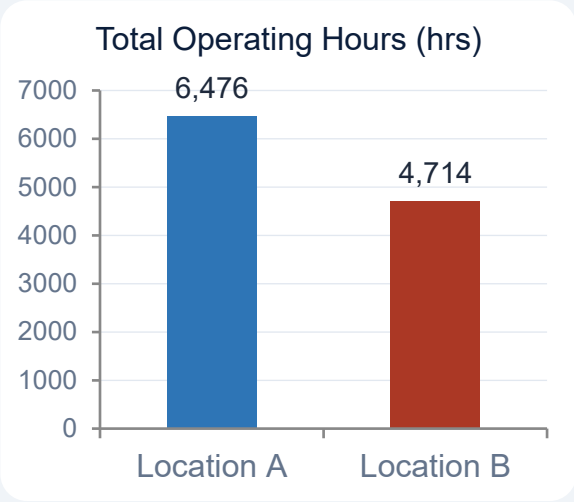
Avg Miles / Intervention

Source: Aurora Fleet Operations Data — Q1 2017 Historical Dataset

VEHICLE UTILIZATION





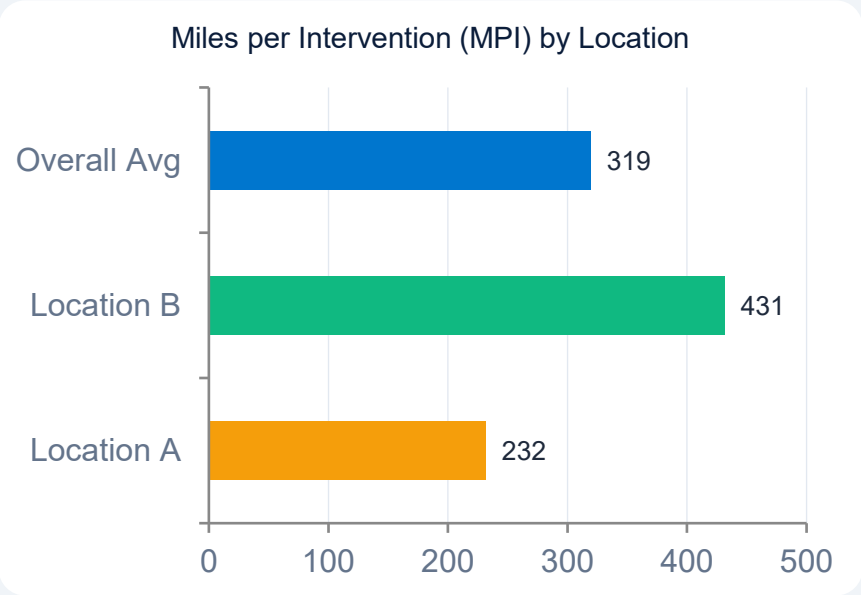


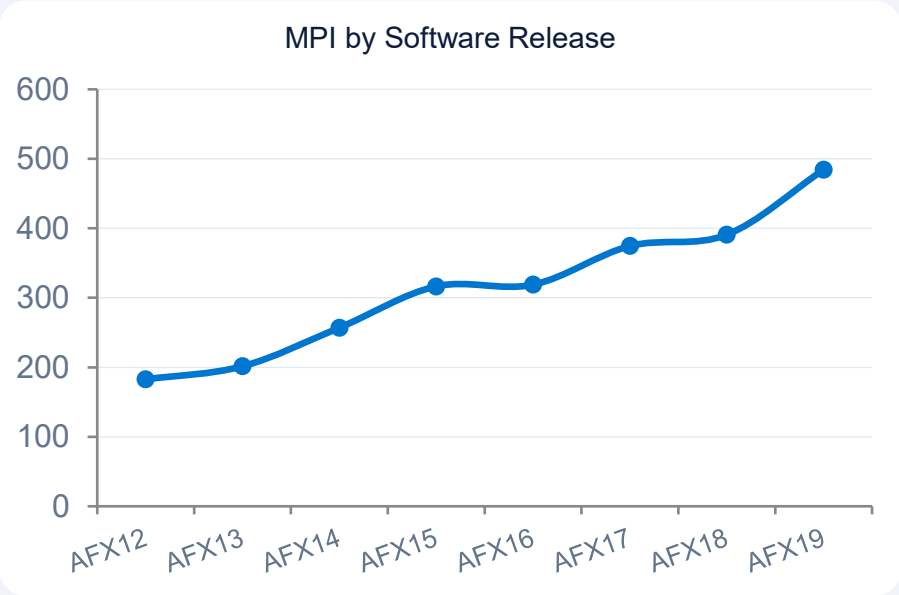
- Location A runs 38% more tests than B, suggesting higher fleet density or demand

- Avg test duration is nearly identical (3.0 hrs) across both locations.
- Setup time averages 12 min (0.2hr) a small but recoverable overhead per mission

- Location A carries 58% of total hours.
- With hours concentrated at location A, OT and cost risk will likely show up at A first during demand spikes.

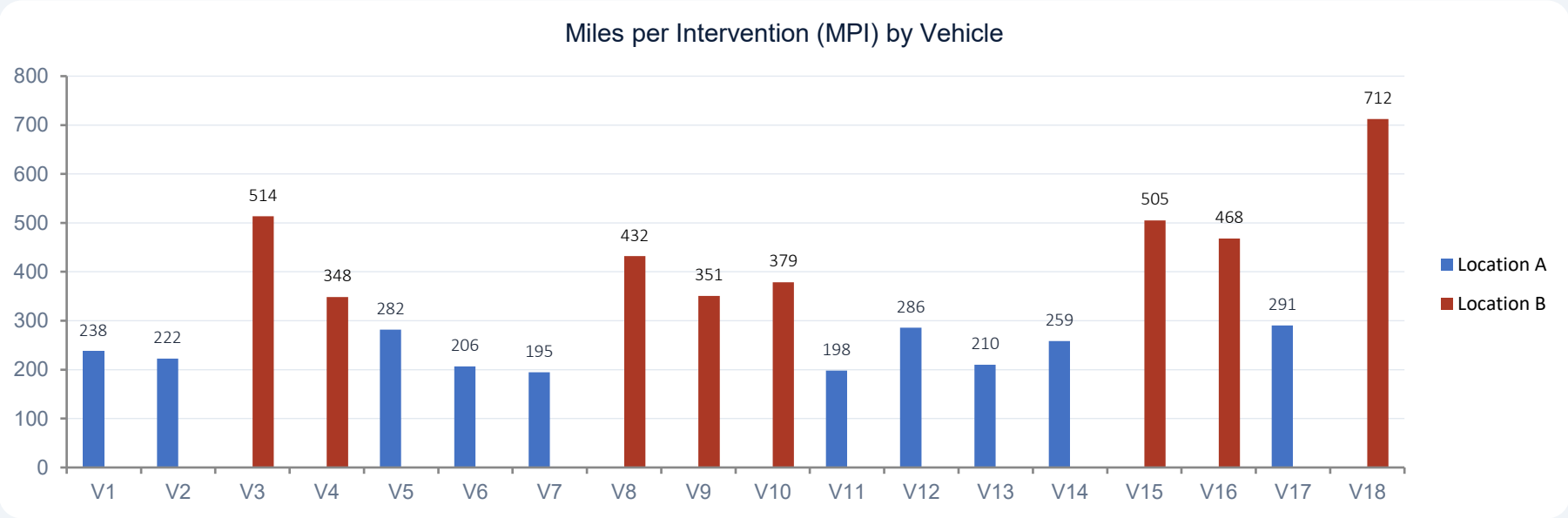
MANUAL INTERVENTION ANALYSIS





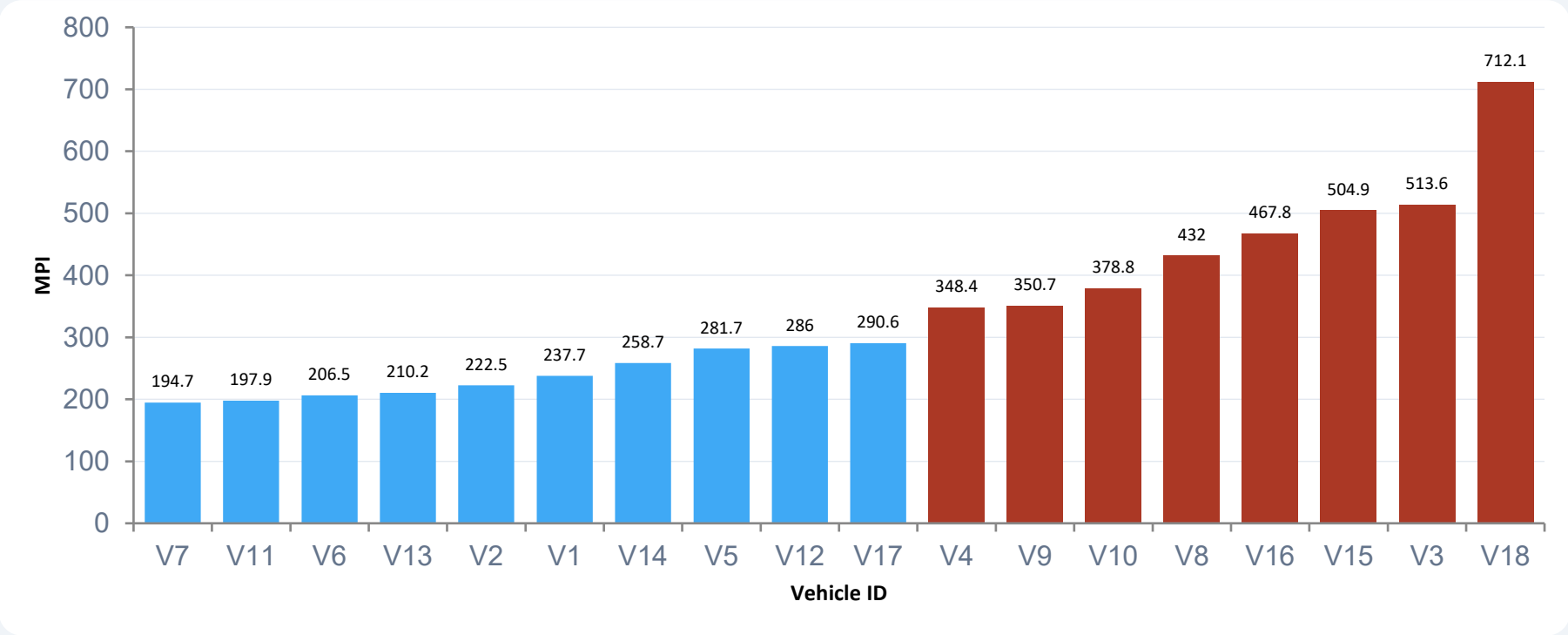
✦ **Key Insight:** Location B's MPI (431 miles) is 86% higher than Location A (232 miles). Software MPI improved **2.65×** from AFX12 to AFX19, demonstrating clear autonomy progression. **MPI will be used as key performance metric for this case.**

MANUAL INTERVENTION ANALYSIS



✦ **Key Insight:** MPI varies 3× across vehicles (195 ~ 712 miles/intervention), so reliability is driven by a small set of low MPI "problem" vehicles. Prioritize these vehicles for triage or maintenance to reduce interventions and unlock higher utilization without adding fleet or headcount.

FLEET HEALTH BY VEHICLE

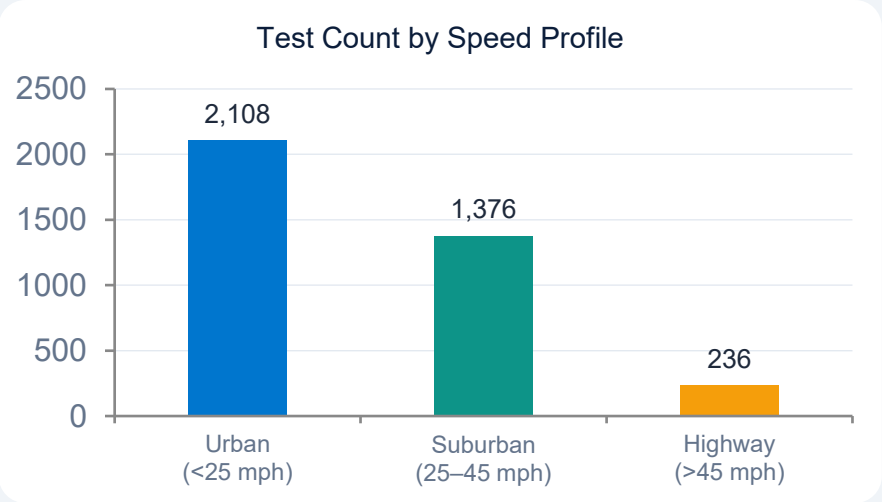


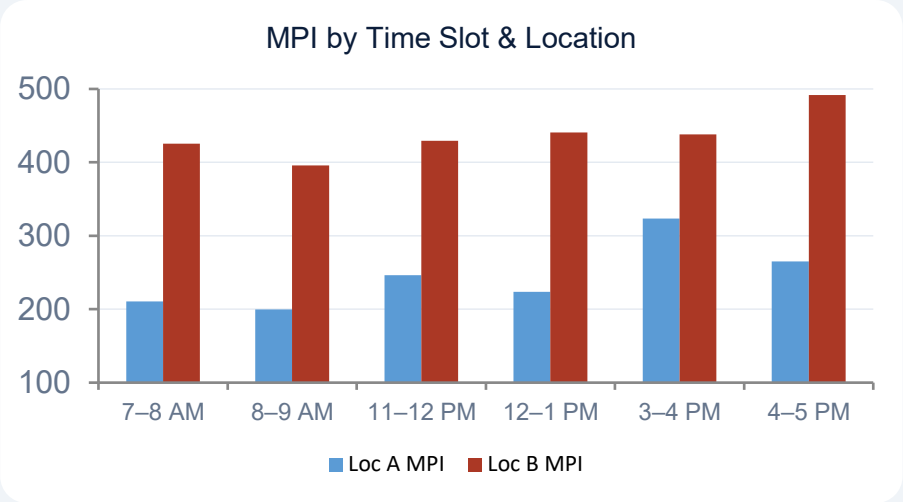
Location A Vehicles

Location B Vehicles

Higher MPI = Fewer interventions = Better performance

SPEED PROFILE & TIME-OF-DAY





Urban (56.7%)

Miles: 124K mi

Loc A dominated 93% urban, only 5.5% suburban

Suburban (37.0%)

Miles: 156K mi

Shared by both locations; highest mile accumulation

Highway (6.3%)

Miles: 37K mi

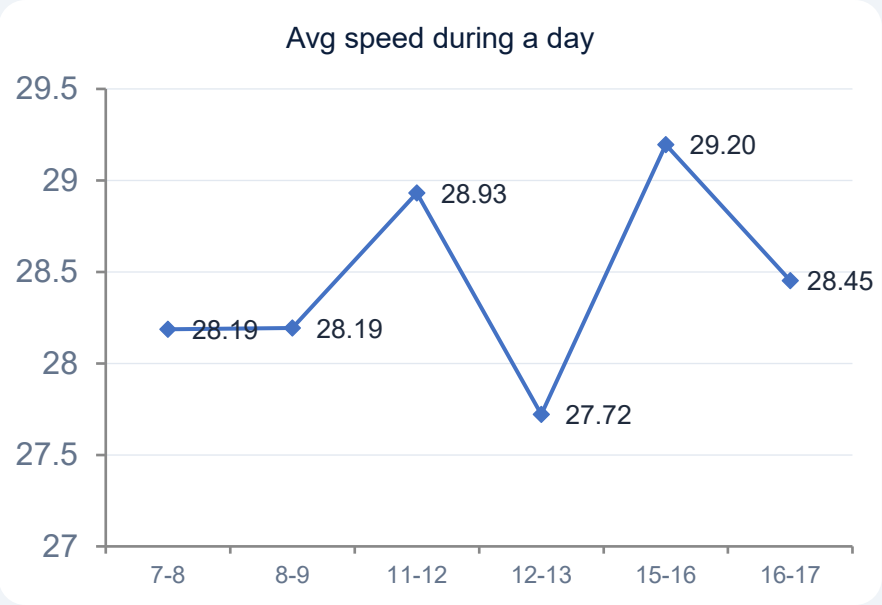
Exclusive to Location B, likely freeway test routes

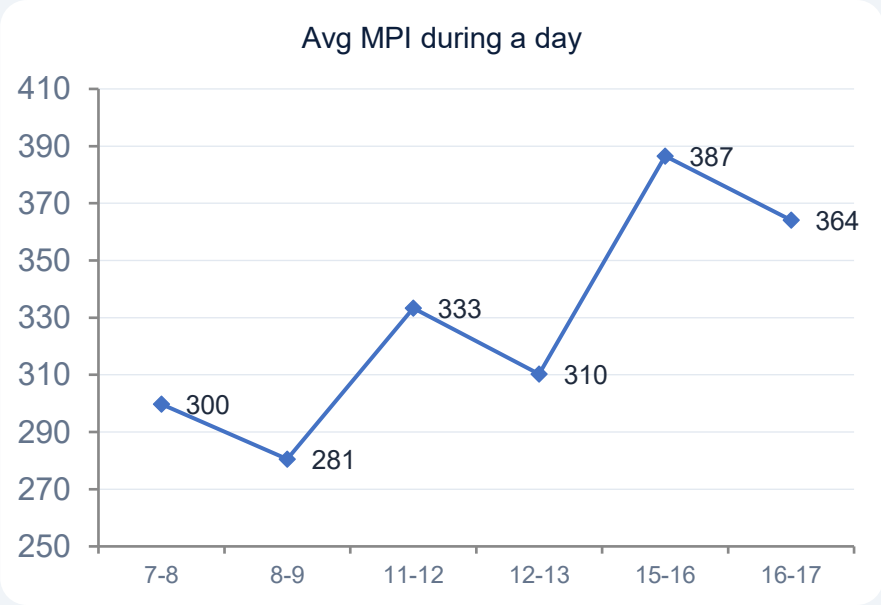
✦ **Insight:** Location B's afternoon slot (4–5 PM) achieves the highest MPI at 491 miles. Afternoon testing slots consistently outperform morning across both sites, a scheduling lever worth exploring. (I'm assigning speed clustering for the categorical group for the purpose of analysis.)

SPEED PROFILE & TIME-OF-DAY

✦ **Insight:** Miles driven is strongly segmented by route context: for the same 3 hours duration, Urban and Suburban trips yield materially less miles than Highway, so planning and forecasting could use route type specific productivity rates rather than a single average.

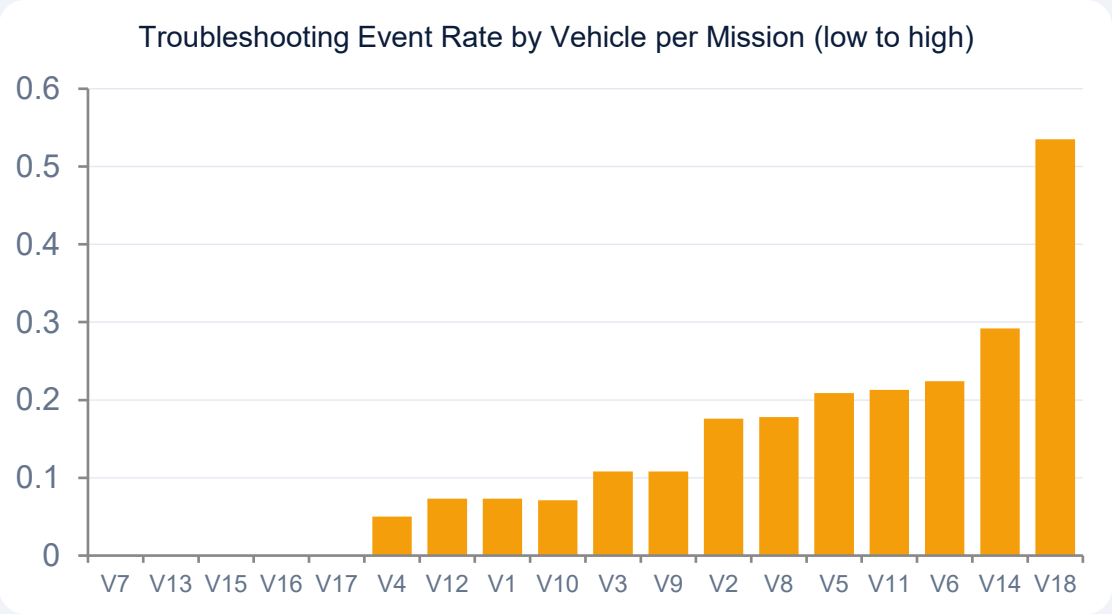
SPEED PROFILE & TIME-OF-DAY





✦ **Insight:** MPI is lowest in the morning and rises into the afternoon (peaking 15–16), while average speed stays nearly flat. This points to reliability being driven less by speed and more by start-of-day stabilization (warm-up/reset effects) and operating conditions (traffic/route complexity). Operationally, treat early hours as a warmup window and shift higher value testing into 15–17 to improve miles per intervention without reducing throughput.

VEHICLE RELIABILITY & TROUBLESHOOTING RATE



RELIABLE (Trouble Rate < 10%)

V7, V13, V17, V12, V1, V4, V10, V15, V16
Zero or near-zero troubleshooting events

MONITOR (10-20% trouble rate)

V3, V9, V2, V8
Moderate troubleshooting frequency

HIGH RISK (>20% trouble rate)

V5, V11, V6, V14, V18
Require inspection/maintenance prioritization

⚠ **Paradox Finding:** V18 has the HIGHEST trouble rate (53.5%) yet the HIGHEST MPI (712 miles/intervention). This suggests V18 may be testing harder routes or scenarios, its interventions are rare but its troubleshooting events are frequent. Warrants separate operational investigation before any retirement decision.

RECOMMENDATIONS

01 Shift Scheduling Optimization

- Increase throughput per vehicle by extending operating window
- upside depends on maintenance + safety constraints.

✓ Up to 2× throughput per vehicle

⚠ Operator fatigue, shift premium costs, maintenance scheduling

02 Vehicle Rotation & Maintenance Strategy

V6, V7 (MPI 195) show significantly higher intervention rates vs top performers like V18 (MPI 712). Triage + targeted maintenance + route reassignment; de-prioritize for critical testing until MPI recovers. If a vehicle remains bottom tile for 2 or 3 consecutive weeks under comparable conditions, it means repair or holdout required

✓ Fewer interventions = faster completion, lower cost per test

⚠ Reduced fleet redundancy; bottleneck if high-MPI vehicles go down

03 Cross-Location Vehicle Rebalancing

Location B achieves 86% higher MPI with 38% fewer tests.

- Pilot rebalancing 1 to 2 vehicles for 2 weeks
- measure MPI lift controlling for route type and time-of-day. If lift persists, scale

✓ Higher system wide MPI, fewer total interventions

⚠ Logistics of transport, crew reallocation, route saturation at Loc B

ADDITIONAL ANALYSES NEEDED

ADDITIONAL DATA WANTED

- Maintenance logs & downtime per vehicle — needed to measure true availability
- Intervention type and cause classification — not all interventions are equal
- Operator-level performance data — to assess human-side variance
- Route and geospatial data — to understand why Location B has higher MPI
- Vehicle age and mileage — to correlate fleet age with performance
- Weather & environmental conditions — external confounders on MPI

ANALYSES TO RUN

- Survival analysis: How does MPI degrade as vehicle mileage accumulates?
- Time-of-day vs MPI regression — are intervention rates correlated with traffic?
- A/B comparison: Structured route testing to isolate location vs route effects
- Predictive maintenance model using troubleshooting event frequency
- Correlation analysis: Software release velocity vs MPI improvement rate
- Operator and vehicle pairing analysis to identify performance outliers

Part II.
Cost-Optimized Headcount Forecast

HEADCOUNT FORECAST — METHODOLOGY

CORE ASSUMPTIONS

Operators per Mission

2

2 operators per mission (driver + safety operator)

Standard Hours/Week

40 hrs

Per operator, regular schedule

OT Rate

1.5×

OT cost modeled at 1.5× (no explicit OT cap in baseline) for hours beyond 40/wk

Out-of-Office Rate

13%

Applied to reduce available hours

Effective Hrs/Operator

34.8 hrs

= 40 × (1 − 13%)

Hourly Rate Assumption

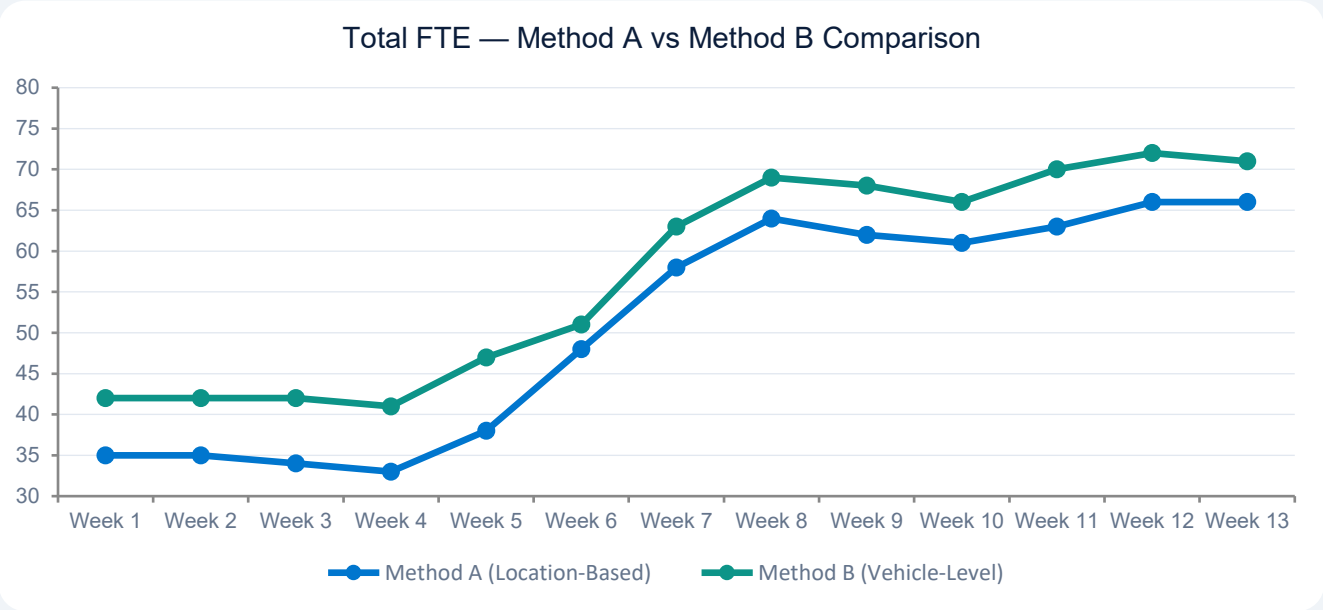
$50/hr

Used in cost optimization model

MODELING APPROACH

1. Aggregate weekly vehicle hours by location and then convert to operator hours (×2 operators/mission)
2. → Apply ETS Time-Series Forecast (Apr 2–Jun 25, ~13 weeks)
3. → Method A (Solver): choose integer HC + OT hours to meet demand at min cost (regular + 1.5× OT)
4. → Method B (crosscheck): forecast weekly hours per vehicle (V1–V18), sum to location total, then translate to HC (same cost logic as Method A)

STAFFING METHODS A VS B - BASELINE



Aurora's current staffing baseline isn't included in the dataset, so I built an estimated "required staffing" baseline from observed/forecasted operator-hours demand and standard capacity assumptions.

Method A (Location-based): converts weekly location demand to FTE = CEILING (Operator Hours/34.8), where 34.8 = 40 × (1–13%))

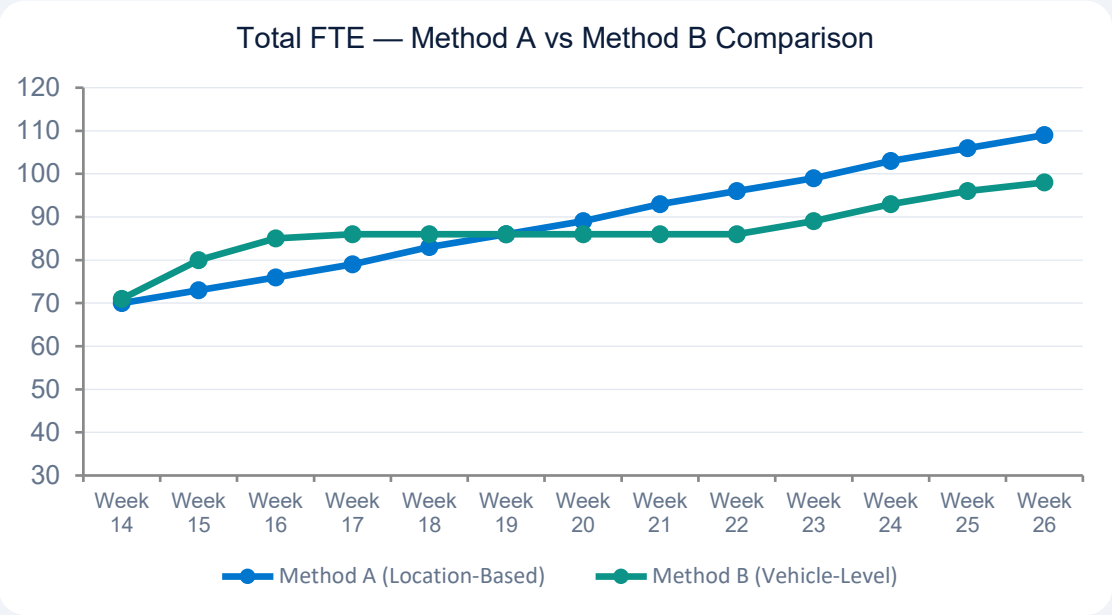
Method B (Vehicle-level crosscheck): forecasts per-vehicle hours (V1–V18), sums to location totals, then applies the same FTE conversion, captures vehicle variability.

Why two methods (what each is good for)

Method A (Location-Based): best for a fast, stable staffing signal, easier to communicate but may understate peaks.
Method B (Vehicle-Level): better crosscheck. captures vehicle dispersion and operational "lumpiness" and tends to produce a safer (higher) staffing need.

Finding: Method B is consistently 5 to 10 FTE higher than Method A.

NON-OT STAFFING & COST FORECAST



Metric	Method A	Method B
Starting FTE	68	68
Peak FTE	109	98
Eff Hrs/Op/Wk	34.8	34.8
Week 1 labor Cost	~$140K	~$124K
Total cost (Apr 2–Jun 25)	~$2.22M	~$1.96M
OT Vehicles		
Granularity	2 buckets	18 vehicles
Best use	Planning	Scheduling

✅ Both align on baseline demand

Same starting FTE; differences emerge as demand ramps, use as a range for planning.

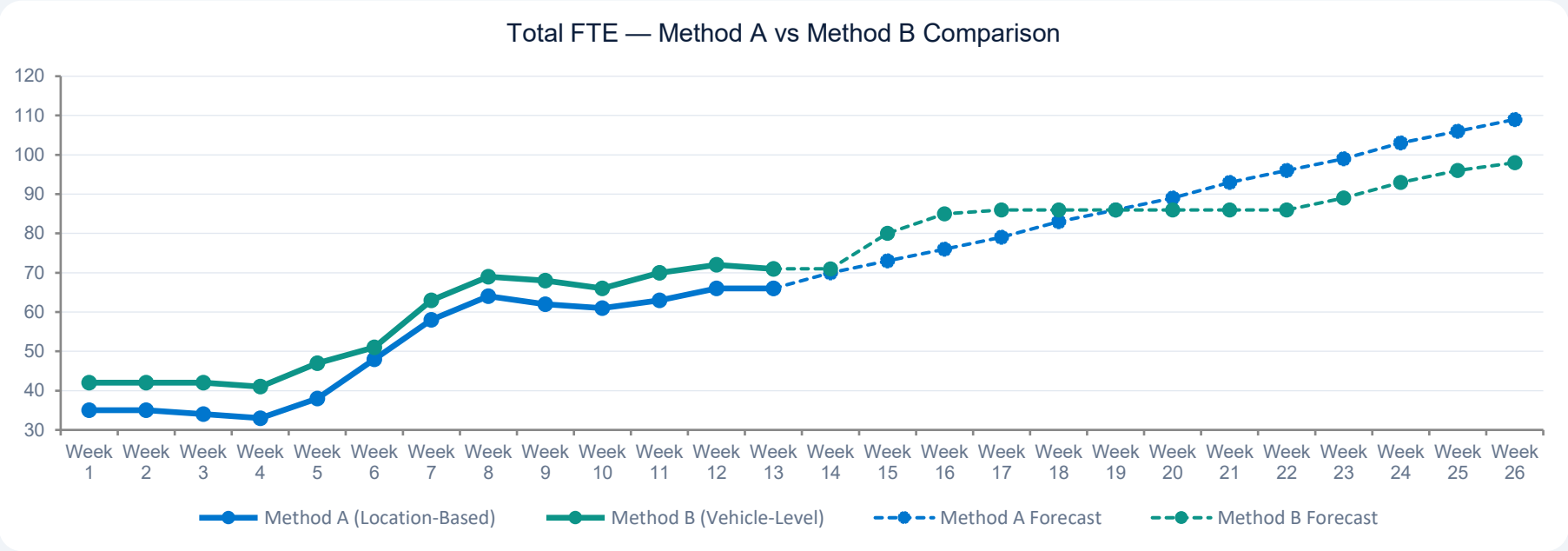
📊 Method A is more conservative in ramp weeks

Location aggregation smooths variability → higher peak HC (+~11 FTE vs Method B).

🎯 Recommendation: Use Method B as the staffing baseline

Under a no-OT constraint, Method B delivers the same coverage logic with lower total cost.

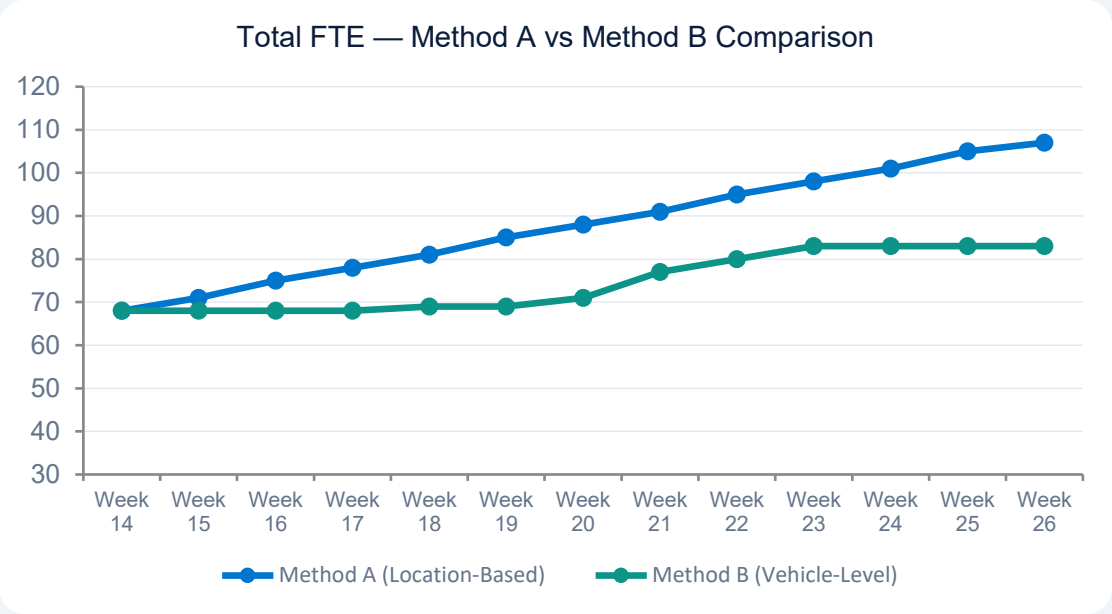
NON-OT STAFFING & COST FORECAST (Apr 2-Jun 25)



Insight

In a non OT senerio, staffing must ramp steadily; Method A becomes the conservative upper bound. By Week 26, Method A forecasts 110 FTE vs 98 FTE for Method B, which is 12 FTE delta driven by location-level smoothing vs vehicle level allocation.

WITH-OT STAFFING & COST FORECAST



Metric	Method A	Method B
Starting FTE	68	68
Peak FTE	107	83
Eff Hrs/Op/Wk	34.8	34.8
Week 1 labor Cost	~$137K	~$121K
Total cost (Apr 2–Jun 25)	~$2.20M	~$1.80M
OT Vehicles	N/A	OT used to absorb peaks
Granularity	2 buckets	18 vehicles
Best use	Planning	Scheduling

✅ OT reduces required headcount, especially in Method B

Vehicle-level scheduling shifts peak load into OT to materially lower peak FTE (83 vs 107)

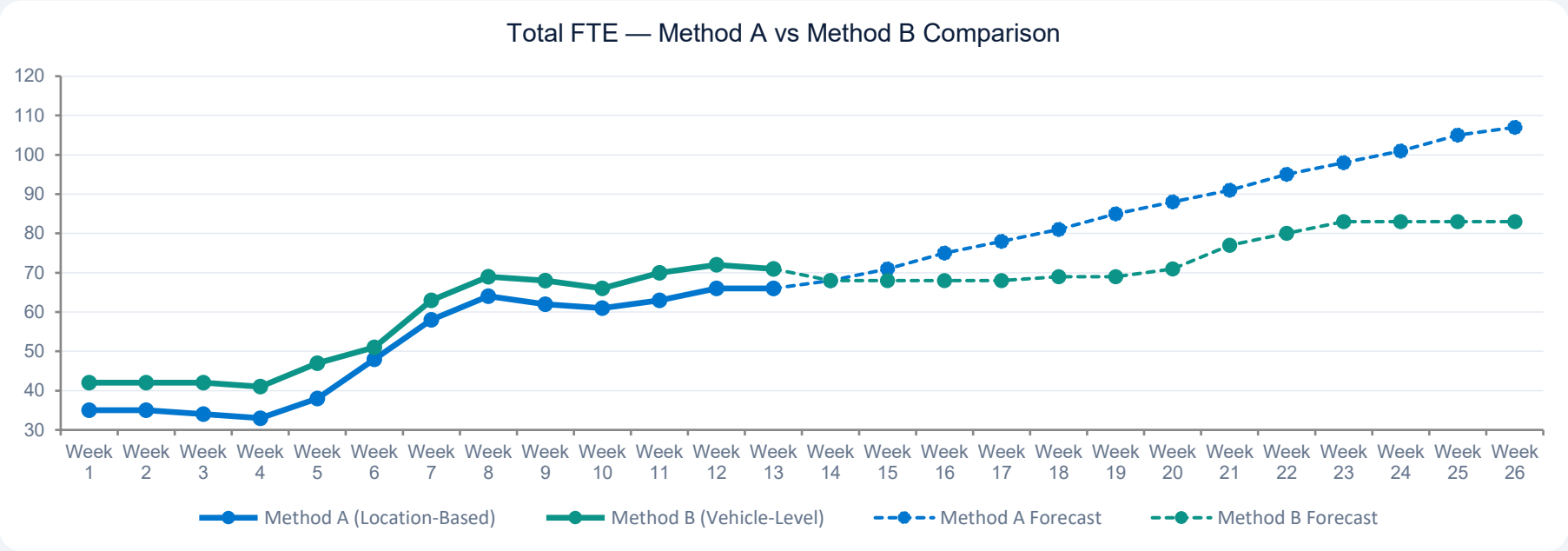
📊 Tradeoff: lower HC vs fatigue / execution risk

Lower total cost depends on OT feasibility, confirm OT limits, safety policy, and sustained workload.

🎯 Recommendation: Present both as decision scenarios

Non-OT = hiring-led plan (higher HC, lower risk).
W/ OT = flexible plan (lower HC, higher policy/people risk).

WITH-OT STAFFING & COST FORECAST (Apr 2-Jun 25)



Insight (WITH OT)

Allowing overtime materially reduces hiring needs, especially under the vehicle-level approach. Method B's forecast flattens around 83 FTE (OT absorbs demand peaks), while Method A still ramps to 107 FTE, implying 24 FTE of "headcount avoided" when OT is used as the flexibility lever. Operationally, this will shift the constraint from hiring speed to OT sustainability (fatigue, coverage quality, and policy limits).

FORECAST METHOD COMPARISON

Method A: Location-Based

ETS forecast by Location + Ceiling(hours/34.8)

✓ Advantages

- Simple to run and explain (2 locations)
- Fast planning baseline for hiring + budgeting
- Stable weekly signal (less noise)
- Works well when vehicle mix is stable

⚠ Limitations

- Aggregation can mask vehicle "lumpiness" and downtime
- Less actionable for dispatching / day-to-day scheduling
- Can over- or under-estimate peak needs depending on vehicle distribution

Starting: 68 FTE | Peak: 109 FTE | 13-Wk: ~$2.22M

Method B: Vehicle-Level Crosscheck

ETS per vehicle (V1–V18) + summed to location + staffing/cost optimization

✓ Advantages

- Reveals which vehicles drive demand and variability
- Better scheduling visibility (supports per-vehicle allocation)
- Enables targeted actions (rebalance fleet / prioritize maintenance)
- More accurate when vehicle availability differs

⚠ Limitations

- Higher data + setup burden (18 vehicles)
- More sensitive to missing data / anomalies
- Requires clearer OT/availability rules to operationalize

Starting: 68 FTE | Peak: 98 FTE | 13-Wk: ~$1.96M

Aurora

Thank You

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INDEX

SUMMARY & NEXT STEPS

Demand is concentrated at Loc A, primary driver of staffing risk

Loc A carries 58% of total operator hours at the same 3 hours test duration, making it the first to face staffing/capacity pressure as demand scales

MPI improvement is the biggest structural cost lever

MPI grew 2.65× from AFX12 to AFX19, each release directly reduces how often operators must intervene, compounding over time

Loc B's MPI advantage likely reflects better route/operating mix

Exclusive highway routes and afternoon peak MPI of 491 mi (4–5 PM) explain the gap; replicating this structure at A is the fastest fleet wide MPI lever

V18: high MPI and high troubleshooting rate

V18 has the highest MPI and highest trouble rate. Treat as a priority diagnostic vehicle: validate reliability drivers before benchmarking or making retirement decisions.

Two method forecast triangulates a staffing range

Location-based forecasting is best for faster planning; vehicle-level forecasting is better for scheduling and identifying outliers. Using both provides a credible staffing band.

RECOMMENDED NEXT STEPS

1. Pilot afternoon shift extension at Loc A (3–5 PM) to capture the peak-MPI window without adding vehicles or hardware

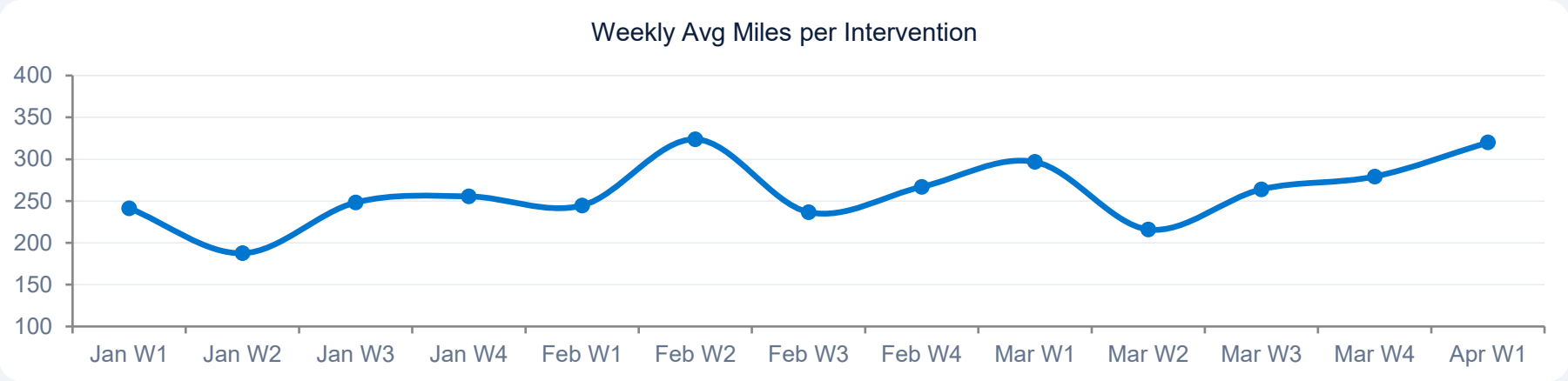
2. Triage V7, V6, V11 (MPI 195–207) with targeted maintenance; track 2 to 3 week MPI recovery under comparable conditions before any retirement decision

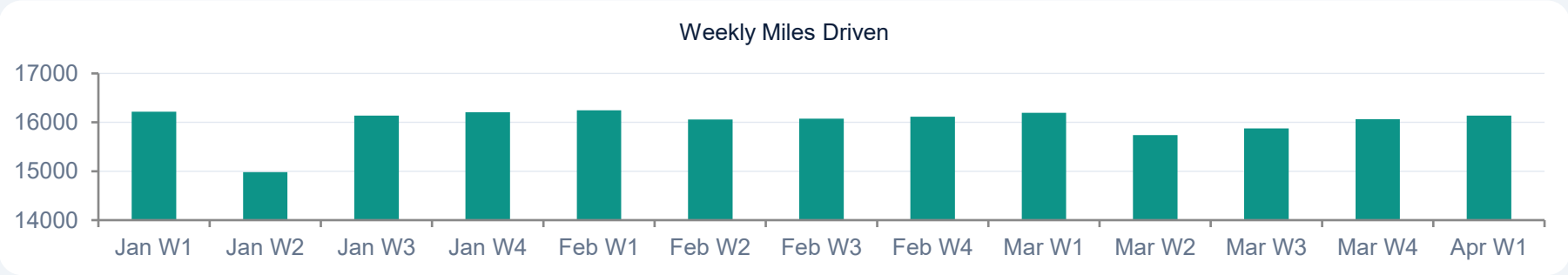
3. Classify all interventions by cause type, an intervention taxonomy is required before software prioritization can be data-driven

4. Confirm OT policy limits and safety thresholds before operationalizing Method B's with OT plan as the scheduling baseline

5. Standardize Loc B's high-MPI route structure at Loc A; measure lift controlling for time of day and vehicle assignment

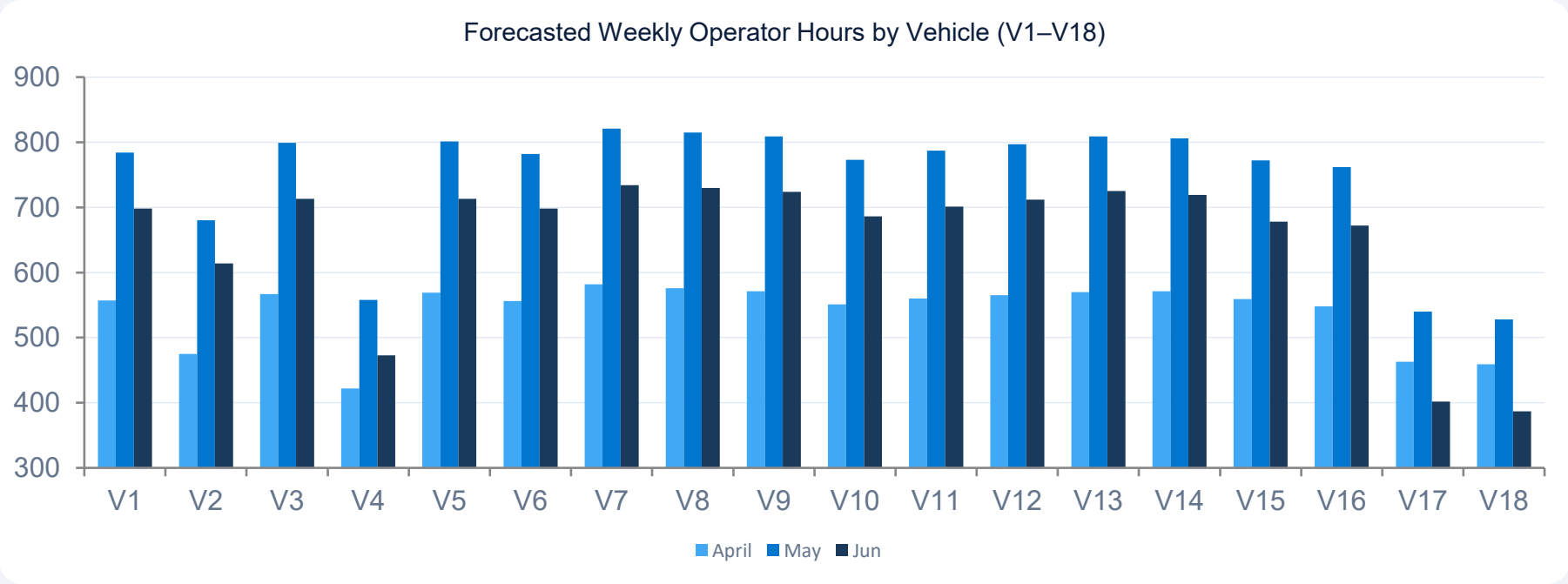
DAILY PERFORMANCE TRENDS (Q1 2017)





Trend: Weekly mileage is very stable (15,700–16,250 mi/week), vehicles run at consistent utilization. MPI is more variable week-to-week, reflecting route mix and software iteration cycles.

VEHICLE-LEVEL OPERATOR HOURS FORECAST (ETS)



Phased availability: V15–V18 start below the main fleet in April → treat as a staged ramp (activation/downtime), not steady-state capacity.

Converging utilization: By June, most vehicles cluster into a similar weekly hour band to demand becomes more evenly distributed across the fleet.

Outliers drive staffing risk: A small set of consistently high-hour vehicles accounts for a disproportionate share of operator-hours, which need schedule rebalancing.